Xinyi (Evelyn) Gao

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Education

University of Chicago **Expected June 2028**
Bachelor of Science in Computer Science and Bachelor of Arts in Statistics (GPA: 3.76/4.0) *Chicago, IL*

Experience

University of Chicago - CHAI Lab **Sep. 2026 – Present**
Undergraduate Researcher, Advisor: Prof. Chenhao Tan *Chicago, IL*

- Study LLM alignment and reasoning reliability, focusing on persuasive misalignment and the potential of causal discovery to improve model reasoning.
- Developed a failure-mode taxonomy and cross-condition analysis framework across three benchmarks, identifying unacknowledged reference dependence, constraint manipulation, and post-hoc rationalization.
- Develop reproducible evaluation experiments to measure how reference answers and user feedback influence model reasoning, extending behavioral analyses into controlled model comparisons.

Salk Institute for Biological Studies **Jun. 2026 – Aug. 2026**
Research Software Engineering Intern, Harnessing Plants Initiative *La Jolla, CA*

- Designed and built bloom-mcp's Python framework for AI-driven scientific analysis, enabling LLM agents to access 23 scientific tools through MCP; established shared validation contracts, typed interfaces, and reproducibility tracking across three repositories.
- Built flexible data infrastructure for scientific workflows, integrating Supabase, S3, and local storage behind unified interfaces; consolidated 649–880 sequential database requests into a single bulk call, eliminating data-loading timeouts.
- Established automated validation for reproducible scientific computing across four experimental platforms, checking numerical outputs against published reference analyses using pytest, Hypothesis property-based tests, and CI gates for determinism, type safety, and serialization.

Sony (China) Limited **May 2025 – Sep. 2025**
Computer Science Operations & Maintenance Intern *Beijing, China*

- Built, deployed, and iteratively refined custom Python automation pipelines tailored to team workflows, processing 100+ weekly music releases across retrieval, metadata cleaning, and delivery; reduced preparation time by ∼60%.
- Designed a configurable feedback-triage tool to meet cross-team coordination needs, routing 300+ weekly comments to 7 teams; separated YAML-based keyword/regex rules from routing logic to support updates without code changes.

Tsinghua University, Prof. Shaojun Zhang's Group **Jul. 2024 – Jan. 2025**
Research Assistant, Spatiotemporal Modeling and Forecasting *Beijing, China*

- Analyzed spatiotemporal vehicle emissions data and developed a 24-hour-ahead forecasting pipeline, benchmarking Random Forest against ARIMA using chronological splits and out-of-sample MAE.
- Identified temporal patterns in forecasting errors and engineered hour-of-day and workday features, reducing MAE through targeted feature refinement while holding the model fixed.

Projects

AlmaBot | *Python, LangGraph, Chroma, MCP* **May 2026 – Aug. 2026**

- Designed a benchmark-driven evaluation framework for an LLM academic advising agent, creating a manually verified 127-question benchmark with fixed calibration/test splits to assess retrieval quality and answer reliability.
- Compared dense, BM25, and hybrid retrieval using paired bootstrap analysis; BM25 exceeded tuned hybrid retrieval by 0.120 MRR@5. Evaluated uncertainty estimation via verbalized confidence and self-consistency, achieving 0.847 AUROC with verbalized confidence and identifying confidence saturation as a limitation for selective abstention.
- Built the advising agent over 4,502 courses across 187 subjects, integrating LangGraph intent routing, course retrieval, and prerequisite-checking tools.

Technical Skills

Languages: Python, SQL, R, Java, C, TypeScript, JavaScript
AI & Machine Learning: LLM evaluation, benchmark design, uncertainty estimation, retrieval evaluation, LangGraph, LangChain, MCP, OpenAI API, scikit-learn
Data & Scientific Computing: NumPy, SciPy, Pandas, GeoPandas, Matplotlib, paired bootstrap analysis, PCA, UMAP, clustering, time-series forecasting
Software Engineering: Git, GitHub Actions, pytest, Hypothesis, Docker, Pydantic, Supabase, React, Next.js